# Denis Sergeev


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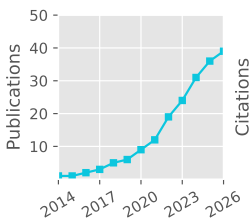


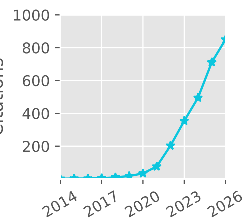


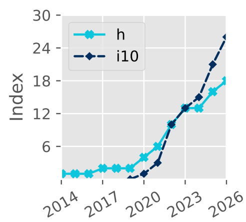


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|---|---|
| Total Pub. | **39** |
| Refereed | **39** |
| First Author | **8** |
| Citations | **850** |
| h-index | **18** |

Updated: 12 Jun 2026

## Career history

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|---|---|
| Jan 2025–now | **Lecturer in Astrophysics**<br>School of Physics, University of Bristol |
| Sep 2021–Dec 2024 | **Postdoctoral Researcher**<br>Project: Exascale Exoplanet Modelling<br>Department of Physics & Astronomy, University of Exeter |
| Sep 2018–Aug 2021 | **Postdoctoral Researcher**<br>Project: Climate Modelling of Rocky Exoplanets<br>Department of Mathematics & Statistics, University of Exeter |

## Academic Qualifications

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|---|---|
| Oct 2014–Aug 2018 | **PhD in Meteorology**<br>Thesis title (shortened): 'Characteristics of Polar Lows in the Nordic Seas'<br>School of Environmental Sciences, University of East Anglia<br>Supervisors: Ian A. Renfrew, Thomas Spengler, Stephen Dorling |
| Sep 2009–May 2014 | **Specialist Diploma (With Honours)**<br>Thesis title: 'Idealised Numerical Modelling of Polar Mesocyclone Dynamics'<br>Department of Meteorology and Climatology, Moscow State University<br>Supervisor: Victor Stepanenko |

## Funding and Awards

| | **Direct Funding, PI** | **Est. Total Value** |
|---|---|---|
| 2026 | Undergraduate Student Bursary \| RAS | £1500 |
| 2026 | Summer Student Placement Bursary \| University of Bristol | ~£4500 |
| 2024 | Above & Beyond Silver Award \| University of Exeter | £1000 |
| 2023 | Meeting Organisation Funding (Exoclimes VI and ExoSLAM) \| RAS | £5000 |
| 2022 | Undergraduate Student Bursary (awarded; student declined) \| RAS | £1200 |
| 2017 | Best Presentation Award \| CEEDA Symposium | ~£100 |
| 2016 | Travel Bursary \| Polar Prediction School | ~£1000 |
| 2015 | Travel Award \| High-Latitude Dynamics workshop | ~£1000 |
| 2014 | Lord Zuckerman PhD scholarship \| School of Environmental Sciences, UEA | ~£112 000 |
| 2014 | Young Scientist Travel Award \| EGU General Assembly | ~£200 |
| 2014 | Russian Academy of Sciences Young Scientist Medal | ~£1000 |
| | **Direct Funding, co-I** | |
| 2025 | Meeting Organisation Funding (UKExoM 2026) \| RAS | £2500 |
| 2025 | Isambard 3 Allocation (30,000 node-hours; PI: N.J. Mayne) \| UKRI (Isambard 3) | ▪ |
| 2024 | UKSA Studentships: Mars Exploration Science | ~£100 000 |
| 2024 | Research Software Engineer Support \| DiRAC HPC | ~£45 000 |
| | **Observational Facilities Resources, co-I** | |
| 2023 | JWST Cycle 2, 49.21 hours (GO 3838, PI: J. Kirk) | ▪ |

## Publications


| # | (preprints in **grey**) | Citations |
|---|---|---|
| 39 | Christie, D. A., Zamyatina, M., Hébrard, E., Evans-Soma, T. M., et al. (incl. **Sergeev, D. E.**), 2026, Benchmarking two chemical networks used in general circulation models of hot Jupiters, MNRAS | ▪ |

38 Lichtenberg, T., Schaefer, L., Krissansen-Totton, J., Miguel, Y., et al. (incl. **Sergeev, D. E.**), 2026, Coupled atmospHere Interior modeL Intercomparison (CHILI)—Protocol Version 1.0: A CUISINES Intercomparison Project of Magma Ocean Models, Planet. Sci. J. **1**

37 Claringbold, A. B., Fisher, C. E., Kirk, J., Ahrer, E., et al. (incl. **Sergeev, D. E.**), 2026, BOWIE-ALIGN: Sub-solar C/O ratio and metallicity atmosphere of the misaligned hot Jupiter HAT-P-30 b, MNRAS **4**

36 Ahrer, E., Fairman, C., Kirk, J., Wakeford, H. R., et al. (incl. **Sergeev, D. E.**), 2025, BOWIE-ALIGN: weak spectral features in KELT-7b's JWST NIRSpec/G395H transmission spectrum imply a high cloud deck or a low-metallicity atmosphere, MNRAS **5**

35 Mak, M. T., **Sergeev, D. E.**, Mayne, N. J., Zamyatina, M., et al., 2025, The impact of different haze types on the atmospheres and observations of hot Jupiters: 3D simulations of HD 189733b, HD 209458b, and WASP-39b, MNRAS **1**

34 **Sergeev, D. E.**, McDermott, J. W., Woods, L., Braam, M., et al., 2025, Lightning activity on a tidally locked terrestrial exoplanet in storm-resolving simulations for a range of surface pressures, MNRAS **1**

33 Meech, A., Claringbold, A. B., Ahrer, E., Kirk, J., et al. (incl. **Sergeev, D. E.**), 2025, BOWIE-ALIGN: substellar metallicity and carbon depletion in the aligned TrES-4b with JWST NIRSpec transmission spectroscopy, MNRAS **13**

32 Kirk, J., Ahrer, E., Claringbold, A. B., Zamyatina, M., et al. (incl. **Sergeev, D. E.**), 2025, BOWIE-ALIGN: JWST reveals hints of planetesimal accretion and complex sulphur chemistry in the atmosphere of the misaligned hot Jupiter WASP-15b, MNRAS **29**

31 Penzlin, A. B. T., Booth, R. A., Kirk, J., Owen, J. E., et al. (incl. **Sergeev, D. E.**), 2024, BOWIE-ALIGN: how formation and migration histories of giant planets impact atmospheric compositions, MNRAS **39**

30 **Sergeev, D. E.**, Boutle, I. A., Lambert, F. H., Mayne, N. J., et al., 2024, The Impact of the Explicit Representation of Convection on the Climate of a Tidally Locked Planet in Global Stretched-mesh Simulations, ApJ **11**

29 Natchiar, S. R. M., Webb, M. J., Lambert, F. H., Vallis, G. K., et al. (incl. **Sergeev, D. E.**), 2024, Reduction in the Tropical High Cloud Fraction in Response to an Indirect Weakening of the Hadley Cell, JAMES **2**

28 Zamyatina, M., Christie, D. A., Hébrard, E., Mayne, N. J., et al. (incl. **Sergeev, D. E.**), 2024, Quenching-driven equatorial depletion and limb asymmetries in hot Jupiter atmospheres: WASP-96b example, MNRAS **21**

27 Mak, M. T., **Sergeev, D. E.**, Mayne, N., Banks, N., et al., 2024, 3D simulations of TRAPPIST-1e with varying $CO_2$, $CH_4$, and haze profiles, MNRAS **10**

26 Villanueva, G. L., Fauchez, T. J., Kofman, V., Alei, E., et al. (incl. **Sergeev, D. E.**), 2024, Modeling Atmospheric Lines by the Exoplanet Community (MALBEC) Version 1.0: A CUISINES Radiative Transfer Intercomparison Project, Planet. Sci. J. **14**

25 Kirk, J., Ahrer, E., Penzlin, A. B. T., Owen, J. E., et al. (incl. **Sergeev, D. E.**), 2024, BOWIE-ALIGN: A JWST comparative survey of aligned versus misaligned hot Jupiters to test the dependence of atmospheric composition on migration history, RAS Techniques and Instruments **25**

24 Mak, M. T., Mayne, N. J., **Sergeev, D. E.**, Manners, J., et al., 2023, 3D Simulations of the Archean Earth Including Photochemical Haze Profiles, J. Geophys. Res.: Atmospheres **10**

23 **Sergeev, D. E.**, Mayne, N. J., Bendall, T., Boutle, I. A., et al., 2023, Simulations of idealised 3D atmospheric flows on terrestrial planets using LFRic-Atmosphere, Geosci. Model Dev. **15**

22 Cohen, M., Bollasina, M. A., **Sergeev, D. E.**, Palmer, P. I., et al., 2023, Traveling Planetary-scale Waves Cause Cloud Variability on Tidally Locked Aquaplanets, Planet. Sci. J. **10**

21 Eager-Nash, J. K., Mayne, N. J., Nicholson, A. E., Prins, J. E., et al. (incl. **Sergeev, D. E.**), 2023, 3D Climate Simulations of the Archean Find That Methane has a Strong Cooling Effect at High Concentrations, J. Geophys. Res.: Atmospheres **8**

20 McCulloch, D., **Sergeev, D. E.**, Mayne, N., Bate, M., et al., 2023, A modern-day Mars climate in the Met Office Unified Model: dry simulations, Geosci. Model Dev. **8**

19 Braam, M., Palmer, P. I., Decin, L., Ridgway, R. J., et al. (incl. **Sergeev, D. E.**), 2022, Lightning-induced chemistry on tidally-locked Earth-like exoplanets, MNRAS **22**

18 Christie, D. A., Lee, E. K. H., Innes, H., Noti, P. A., et al. (incl. **Sergeev, D. E.**), 2022, CAMEMBERT: A Mini-Neptunes General Circulation Model Intercomparison, Protocol Version 1.0.A CUISINES Model Intercomparison Project, Planet. Sci. J. **9**

17 **Sergeev, D. E.**, Fauchez, T. J., Turbet, M., Boutle, I. A., et al., 2022, The TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI). II. Moist Cases-The Two Waterworlds, Planet. Sci. J. **77**

16 Turbet, M., Fauchez, T. J., **Sergeev, D. E.**, Boutle, I. A., et al., 2022, The TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI). I. Dry Cases-The Fellowship of the GCMs, Planet. Sci. J. **63**

15 Fauchez, T. J., Villanueva, G. L., **Sergeev, D. E.**, Turbet, M., et al., 2022, The TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI). III. Simulated Observables-the Return of the Spectrum, Planet. Sci. J. **55**

14 **Sergeev, D. E.**, Lewis, N. T., Lambert, F. H., Mayne, N. J., et al., 2022, Bistability of the Atmospheric Circulation on TRAPPIST-1e, Planet. Sci. J. **33**

13 Cohen, M., Bollasina, M. A., Palmer, P. I., **Sergeev, D. E.**, et al., 2022, Longitudinally Asymmetric Stratospheric Oscillation on a Tidally Locked Exoplanet, ApJ **17**

12 Fauchez, T. J., Turbet, M., **Sergeev, D. E.**, Mayne, N. J., et al., 2021, TRAPPIST Habitable Atmosphere Intercomparison (THAI) Workshop Report, Planet. Sci. J. **39**

11 Terpstra, A., Renfrew, I. A., & **Sergeev, D. E.**, 2021, Characteristics of Cold-Air Outbreak Events and Associated Polar Mesoscale Cyclogenesis over the North Atlantic Region, J. Cli. **29**

10 Renfrew, I. A., Barrell, C., Elvidge, A. D., Brooke, J. K., et al. (incl. **Sergeev, D.**), 2021, An evaluation of surface meteorology and fluxes over the Iceland and Greenland Seas in ERA5 reanalysis: The impact of sea ice distribution, Q. J. R. Meteorol. Soc. **74**

9 Eager-Nash, J. K., Reichelt, D. J., Mayne, N. J., Hugo Lambert, F., et al. (incl. **Sergeev, D. E.**), 2020, Implications of different stellar spectra for the climate of tidally locked Earth-like exoplanets, A&A **27**

8 **Sergeev, D. E.**, Lambert, F. H., Mayne, N. J., Boutle, I. A., et al., 2020, Atmospheric Convection Plays a Key Role in the Climate of Tidally Locked Terrestrial Exoplanets: Insights from High-resolution Simulations, ApJ **67**

7 Joshi, M. M., Elvidge, A. D., Wordsworth, R., & **Sergeev, D.**, 2020, Earth's Polar Night Boundary Layer as an Analog for Dark Side Inversions on Synchronously Rotating Terrestrial Exoplanets, ApJ **20**

6 Renfrew, I. A., Pickart, R. S., Våge, K., Moore, G. W. K., et al. (incl. **Sergeev, D.**), 2019, The Iceland Greenland Seas Project, BAMS **28**

5 **Sergeev, D.**, Renfrew, I. A., & Spengler, T., 2018, Modification of Polar Low Development by Orography and Sea Ice, Mon. Wea. Rev. **18**

4 Shestakova, A. A., Toropov, P. A., Stepanenko, V. M., **Sergeev, D. E.**, et al., 2018, Observations and modelling of downslope windstorm in Novorossiysk, Dyn. Atm. Ocean. **7**

3 **Sergeev, D. E.**, Renfrew, I. A., Spengler, T., & Dorling, S. R., 2017, Structure of a shear-line polar low, Q. J. R. Meteorol. Soc. **22**

2 Spengler, T., Renfrew, I. A., Terpstra, A., Tjernström, M., et al. (incl. **Sergeev, D.**), 2016, High-Latitude Dynamics of Atmosphere-Ice-Ocean Interactions, BAMS **7**

1 Eliseev, A. V., & **Sergeev, D. E.**, 2014, Impact of subgrid-scale vegetation heterogeneity on the simulation of carbon-cycle characteristics, Izv. Atmos. Ocean. Phy. **9**


## Conferences and Seminars

### Invited Talks

| | |
|---|---|
| May 2025 | General circulation models for exoplanet atmospheres<br>JpGU-AGU Joint Meeting 2026 \| Chiba, Japan |
| Oct 2025 | CUISINES - a framework for exoplanet model intercomparison projects<br>Atmospheric and interior evolution of planetary magma oceans \| Leiden, the Netherlands |
| Jun 2025 | Atmospheric dynamics on other planets<br>Durham HPC Days \| Durham, UK |
| Feb 2025 | Exoplanet climate modelling with LFRic<br>University of East Anglia \| Norwich, UK |
| May 2024 | 3D simulations of exoplanet atmospheres with the next-generation Met Office model<br>University of Leicester \| Leicester, UK |
| Apr 2024 | Shall I compare thee to a distant world? Inter-planet and inter-model comparative studies<br>EGU General Assembly \| Vienna, Austria |

Jul 2023 Simulations of idealised 3D atmospheric flows on terrestrial planets using LFRic-Atmosphere
NASA GISS Seminar | Online

Mar 2023 First results of using LFRic for exoplanet climate modelling
NIWA Seminar | Wellington, New Zealand

Feb 2023 Atmospheric dynamics and chemistry on exoplanets
UQ Astro Group Meeting | Brisbane, Australia

Feb 2023 Atmospheric dynamics and chemistry on exoplanets
UniSQ Exoplanet Group Seminar | Brisbane, Australia

Feb 2023 Atmospheric dynamics and chemistry on exoplanets
UNSW AstroSeminar | Sydney, Australia

Apr 2022 Dichotomy of the atmospheric circulation on TRAPPIST-1e
NASA GISS Seminar | Online

Jan 2022 Dichotomy of the atmospheric circulation on TRAPPIST-1e
NASA GSFC Extrasolar Planets Seminar | Online

Nov 2021 TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI)
MPIA APEx Exocoffee | Online

May 2021 Overcast on TRAPPIST-1e
RCC MSU Geophysical Seminar | Online

Sep 2020 Simulations of convection over a range of atmospheric conditions on TRAPPIST-1e
THAI Workshop | Online

Apr 2020 Atmospheric convection plays a key role in the climate of tidally locked exoplanets
University of Reading Meteorology Seminar | Online

Apr 2020 Atmospheric convection plays a key role in the climate of tidally locked exoplanets
NASA GISS Seminar | Online

## Contributed Talks

May 2026 Lightning climatology on TRAPPIST-1e
ELSI workshop | Tokyo, Japan

Sep 2023 Introducing GeoVista - Cartographic rendering and mesh analytics powered by PyVista (joint talk)
Met Office Seminar | Exeter, UK

Jul 2022 Bistability of the atmospheric circulation on TRAPPIST-1e
Rocky Worlds II | Oxford, UK

Apr 2022 Dichotomy of the atmospheric circulation on TRAPPIST-1e
Exoplanet Modelling in the James Webb Era II: Terrestrial planets and sub-Neptunes | Online

Nov 2020 Explicit convection on tidally locked rocky exoplanets simulated with the UM nesting suite
Unified Model users workshop | Online

Aug 2019 Simulations of moist convection on tidally-locked rocky exoplanets
Exoclimes V | Oxford, UK

Jun 2019 North Atlantic polar mesoscale cyclones in ERA5 and ERA-Interim reanalyses
IGP workshop | Norwich, UK

Apr 2019 Atmospheric convection on tidally-locked Earth-like exoplanets
UK Exoplanet Community Meeting | London, UK

Jun 2018 Modification of Polar Low Development by Sea Ice and Svalbard Orography
POLAR2018 | Davos, Switzerland

Oct 2017 The influence of Svalbard orography and sea ice on polar low development
18th Cyclone Workshop | Sainte-Adèle, Canada

Apr 2017 Polar lows and how background environment can influence their development
Cambridge Earth Systems Science EnvEast Doctoral Alliance Symposium | Cambridge, UK

May 2016 Structure of the shear-line polar low south of Svalbard
NORPAN meeting | Tokyo, Japan

Apr 2016 Structure of the shear-line polar low south of Svalbard
13th European Polar Lows Working Group Workshop | Paris, France

## Poster Presentations

Nov 2025 Lightning climatology on rocky exoplanets in a global storm-resolving model
CTR Wilson Meeting on Atmospheric Electricity | Bath, UK
Jun 2024 The impact of convection on the climate of a tidally locked planet in stretched-mesh simulations
Exoplanets 5 | Leiden, Netherlands
Apr 2024 The impact of convection on the climate of TRAPPIST-1e in global stretched-mesh simulations
EGU General Assembly | Vienna, Austria
Apr 2024 The impact of convection on the climate of a tidally locked planet in stretched-mesh simulations
UK Exoplanet Community Meeting | Birmingham, UK
Nov 2022 Dry Modern-Day Mars Climate in the Met Office Unified Model
UK Solar System Planetary Atmospheres | London, UK
Sep 2022 Bistability of the Atmospheric Circulation on TRAPPIST-1e
UK Exoplanet Community Meeting | Edinburgh, UK
Jul 2015 Structure and dynamics of a shear-line polar low during a cold-air outbreak over the Norwegian Sea
Royal Meteorological Society Student Conference | Birmingham, UK
Mar 2015 Structure and dynamics of a shear-line polar low during a cold-air outbreak over the Norwegian Sea
Dynamics of Atmosphere-Ice-Ocean Interactions in the High Latitudes workshop | Rosendal, Norway
May 2014 Numerical modelling of polar mesocyclones dynamics diagnosed by the energy budget
EGU General Assembly | Vienna, Austria
Apr 2013 Impact of subgrid-scale vegetation heterogeneity on the carbon cycle
EGU General Assembly | Vienna, Austria
Apr 2013 Numerical modelling of polar mesocyclones generation mechanisms
EGU General Assembly | Vienna, Austria

## Supervision

(Projects with me as the lead supervisor are in **bold**. Students who continued their academic career are underlined.)

### PhD Supervision

Sep 2025–Sep 2029 **Alex Corbett** (U. Bristol)
**Project: Convection on Sub-Neptunes**
Co-supervisors: B. Shipway, Z. Leinhardt
Sep 2025–Sep 2029 Will Luscombe
Project: Forecasting Martian dust storms
Co-supervisors: N. J. Mayne, M. Bate, B. Drummond
Sep 2021–Apr 2025 Mei Ting (Martha) Mak (U. Exeter)
Project: Hazes in Planetary Atmospheres
Co-supervisors: N. J. Mayne, J. Manners, E. Hébrard

### MSc Supervision

Sep 2025–May 2026 **Evie Cushing**
Project: **Ice Giant Atmospheric Modelling**
Sep 2025–May 2026 **Freya Evans & Daisy Green**
Project: **Atmospheric Dynamics on Ice Giants**
Sep 2025–May 2026 **Catherine Kerr & Lily Odhuba**
Project: **Lightning Storms on Earth-like Exoplanets**
Jan 2023–May 2024 Tom Batchelor, Luke Benzing, & Alex McGinty
Project: Mars Atmosphere Modelling
Co-supervisors: M. Bate, N. J. Mayne, D. McCulloch
Sep 2020–Sep 2022 Danny McCulloch (MSci by Research)
Project: Climate Modelling of Modern-Day Mars
Co-supervisors: M. Bate, N. J. Mayne
Apr 2021–Sep 2022 Meghan Plumridge (MSci by Research)
Project: Climate Modelling of Early Mars
Co-supervisors: M. Bate, N. J. Mayne
Jan 2021–May 2022 Jasper Chadwick & Esse Sellwood

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| | Project: Ocean Heat Transport on Rocky Exoplanets<br>Co-supervisors: F. H. Lambert, J. Eager-Nash |
| Jan 2021–May 2022 | Isabelle Browne & Oakley Young<br>Project: Greenhouse Effect on Early Mars<br>Co-supervisors: F. H. Lambert, N. J. Mayne, J. Eager-Nash |
| Jan 2020–May 2021 | Toby Ferrison<br>Project: Titan Climate Modelling<br>Co-supervisor: F. H. Lambert |
| Oct 2018–May 2019 | Jake Eager-Nash & David Reichelt<br>Project: Implications of Stellar Type on the Climate of Tidally Locked Terrestrial Exoplanets<br>Co-supervisors: F. H. Lambert, N. J. Mayne |

### Undergraduate and Summer Internship Supervision

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|---|---|
| Jul–Sep 2022 | Oakley Young<br>Project: Ekman Ocean Model<br>Co-supervisors: J. Eager-Nash, F. H. Lambert |
| Jun–Sep 2022 | **James McDermott & Lottie Woods**<br>Project: **Simulations of Lightning Storms on Tidally Locked Rocky Exoplanets** |
| Jun–Aug 2021 | Oakley Young<br>Project: Climate Modelling of Archean Earth<br>Co-supervisors: J. Eager-Nash, N. J. Mayne |
| Jun–Aug 2021 | Joshua Parkin & Esse Sellwood<br>Project: The Impact of Host Star Spectrum on the Climate of Rocky Exoplanets<br>Co-supervisors: J. Eager-Nash, N. J. Mayne |
| Jun–Aug 2019 | Isobel Parry<br>Project: Water Cycle on Proxima Centauri b<br>Co-supervisor: F. H. Lambert |

## PhD Examinations

2025 Maria Di Paolo | University of East Anglia | Supervisors: Manoj Joshi, David Stevens

## Teaching and Mentoring

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|---|---|
| 2026–now | Environmental Physics<br>Lecturer \| University of Bristol \| ∼40 students |
| 2025–now | Practical Physics III: Research Skills and Group Project<br>Tutor \| University of Bristol \| 2 groups of ∼7 students |
| 2025–now | Research Project in Physics<br>Supervisor & assessor \| University of Bristol \| ∼10 students |
| Jul 2024 | Algorithms For Exascale Summer School<br>Invited lecturer \| University of Exeter \| ∼20 students |
| Feb 2024 | Physics of Climate Change<br>Workshop lead \| University of Exeter \| ∼30 students |
| Jul 2023 | Climatematch Academy<br>Mentor \| Online \| 3 groups of ∼5 students |
| Jul 2023 | International Sustainability Summer School<br>Lecturer \| University of Exeter \| ∼10 students |
| Jun 2023 | Exoclimes Summer School in Atmospheres and Modelling (ExoSLAM)<br>Lecturer \| University of Exeter \| ∼50 students |
| 2016–2018 | Introduction to Python in Environmental Sciences<br>Course creator & lead \| University of East Anglia \| ∼50 students |
| 2015–2017 | Modelling Environmental Processes; Meteorology; Numerical Skills<br>Teaching assistant \| University of East Anglia |

## Research Leadership and Impact

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|---|---|
| 2024–now | Co-lead of Climates Using Interactive Suites of Intercomparisons Nested for Exoplanet Studies (CUISINES) |
| Jun 2023 | Co-chair of Exoclimes Summer School in Atmospheres and Modelling (ExoSLAM) |
| 2023 | Interview by the University of Exeter about my research |
| 2023 | Interview by UKRI/STFC about my outreach |
| 2023 | Expert Scientist at the British Science Festival Climate Exhibition |
| 2022 | Press releases: University of Exeter, American University, & INSU CNRS |
| 2020–now | 3D visualisations of exoplanet simulations:<br>'Cloudy Skies of Distant Exoplanets' \| University of Exeter Images of Research 2023<br>'A refined look at tidally locked exoplanets' \| DiRAC HPC Research Image Competition 2023<br>'Exoplanetary Atmospheres' \| Exeter Science Centre, Science as Art Gallery 2020<br>'Dusty exoplanet atmospheres' \| Nature Press Release<br>'Virtual Reality Exploration of Exoplanets' \| 360 VR video (contributor) |
| 2019 | Science consulting on the 'Exoplanet Explorers' videogame |
| 2015 | Blogging:<br>Disastrous Disaster Movies<br>Polar Lows: What Fuels Arctic Hurricanes?<br>Worldwide Weird Weather Words |

## Organisation of Scientific Meetings

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|---|---|
| Jun 2026 | Idealised modelling with LFRic: training course (Co-Chair and Instructor) \| Exeter, UK \| ~50 attendees |
| Mar 2026 | UK Exoplanet Community Meeting (SOC) \| Bristol, UK |
| Oct 2025 | Atmospheric and interior evolution of planetary magma oceans (SOC) \| Leiden, the Netherlands |
| Sep 2025 | BUFFET-5 (Co-chair) \| Bordeaux, France |
| Jul 2025 | Exoclimes VII (SOC) \| Montreal, Canada |
| Jun 2025 | Idealised modelling with LFRic (Chair) \| Exeter, UK \| ~50 attendees |
| Oct 2024 | BUFFET-4: Building a Unified Framework For Exoplanet Treatments (Co-chair) \| Online |
| Jun 2024 | What's Cookin' Doc? A CUISINES meeting (Chair) \| Leiden, the Netherlands \| ~20 attendees |
| Jun 2023 | ExoSLAM Summer School (Co-chair) \| Exeter, UK \| ~50 attendees |
| Jun 2023 | Exoclimes VI (LOC) \| Exeter, UK \| ~200 attendees |
| Mar 2023 | Challenge of Science Leadership Short Course \| Exeter, UK |
| Aug 2022 | Exoclimatology Theory Group Summer Retreat \| Mawgan Porth, UK \| 15 attendees |

## Reviewing and Academic Service

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|---|---|
| Journals | Nat. Astron., MNRAS, Planet. Sci. J., Geophys. Res. Lett., ApJ, Planet. Space Sci., Q. J. R. Meteorol. Soc. |
| Funding | STFC Consolidated Grant, STFC ERF, NASA XRP, STFC Small Award |
| Observations | James Webb Space Telescope General Observer Programs (Exoplanets & Disks, Cycles 3 & 4) |
| Membership | Royal Astronomical Society, Europlanet Society |

## Technical Skills

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|---|---|
| Numerical models | LFRic, Unified Model, SOCRATES, LAGRANTO, Isca |
| Programming languages | Python, FORTRAN, MATLAB, NCL |
| Python libraries (user) | cartopy, cython, iris, matplotlib, numpy, pandas, pyvista, xarray |
| Python libraries (creator/contributor) | aeolus, cartopy, pyvista, geovista |
| Parallel computing | Dask, MPI, OpenMP |
| Version control | Git, Subversion |
| Document preparation | LaTeX, Quarto, Jupyter Notebooks, Markdown, HTML, CSS, reST |

## Vocational Training

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|---|---|
| Sep 2023 | Belbin Training |
| Mar 2023 | Challenge of Science Leadership |
| Dec 2022 | Interview Training |
| Jul 2020 | Writing Workshop for Climate Scientists |
| Mar 2020 | ESA JWST Master Class |
| Jul 2019 | ICTP Summer School on Convective Organization and Climate Sensitivity |
| Apr 2018 | Fortran Modernisation Workshop |
| Jan 2018 | Helicopter Underwater Escape Training Course (CA-EBS) |
| Dec 2017 | Sea Survival Course |
| Jun 2017 | Weather Presenting |
| Feb 2017 | Level 1 First Aid for Field Work Course |
| Jan 2017 | Raspberry Pi Basics |
| Apr 2016 | WWRP/WCRP/Bolin Center Polar Prediction School |
| Dec 2014 | UK Met Office Unified Model Training |

## Vocational Experience

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|---|---|
| Apr–Jun 2018 | Data Technician<br>Processing of meteorological data collected in the IGP field campaign \| University of East Anglia |
| 2015–2018 | Founder and Leader<br>Python Users Group \| University of East Anglia |
| Feb–Mar 2018 | Member of the Meteorology Team<br>The Iceland-Greenland Seas Project (IGP) field campaign \| Akureyri, Iceland |
| Mar 2015 | Rapporteur<br>Dynamics of Atmosphere-Ice-Ocean Interactions in the High-Latitudes \| Rosendal, Norway |
| Oct 2013 | Research Intern<br>Geophysical Institute \| University of Bergen, Norway |
| Aug–Sep 2013 | Trainee Forecaster<br>Forecast and Briefing Service \| Main Aviation Meteorological Centre, Vnukovo Airport |
| Jul 2012 | Research Intern<br>A.M. Obukhov Institute of Atmospheric Physics \| Moscow, Russia |

## Unsuccessful Funding Applications

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|---|---|
| 2026 | GW4 Building Communities Generator Fund |
| 2025 | STFC Small Award |
| 2025 | Leverhulme Research Leadership Award |
| 2025 | ERC Starting Grant |
| 2025 | 2nd GPU Embedded CSE (eCSE) support call |
| 2024 | 1st GPU Embedded CSE (eCSE) support call |
| 2023 | STFC Ernest Rutherford Fellowship |
| 2021 | Leverhulme Trust Early Career Fellowship |